

Technical training.
Product information.

F34 Complete Vehicle



BMW Service

Edited for the U.S. market by:
BMW Group University
Technical Training

ST1304

6/1/2013

General information

Symbols used

The following symbol is used in this document to facilitate better comprehension or to draw attention to very important information:



Contains important safety information and information that needs to be observed strictly in order to guarantee the smooth operation of the system.

Information status and national-market versions

BMW Group vehicles meet the requirements of the highest safety and quality standards. Changes in requirements for environmental protection, customer benefits and design render necessary continuous development of systems and components. Consequently, there may be discrepancies between the contents of this document and the vehicles available in the training course.

This document basically relates to the European version of left-hand drive vehicles. Some operating elements or components are arranged differently in right-hand drive vehicles than shown in the graphics in this document. Further differences may arise as a result of the equipment specification in specific markets or countries.

Additional sources of information

Further information on the individual topics can be found in the following:

- Owner's Handbook
- Integrated Service Technical Application.

Contact: conceptinfo@bmw.de

©2013 BMW AG, Munich

Reprints of this publication or its parts require the written approval of BMW AG, Munich

The information contained in this document forms an integral part of the technical training of the BMW Group and is intended for the trainer and participants in the seminar. Refer to the latest relevant information systems of the BMW Group for any changes/additions to the technical data.

Information status: **March 2013**
BV-72/Technical Training

F34 Complete Vehicle

Contents.

1.	Introduction.....	1
1.1.	Models.....	2
1.2.	Performance.....	2
1.3.	Dimensions.....	3
1.4.	Weights and payload.....	3
1.5.	Silhouette comparison.....	4
2.	Bodyshell.....	5
3.	Doors.....	6
3.1.	Inner door panel/Door outer skin.....	6
3.2.	Side windows.....	7
4.	Tailgate.....	9
4.1.	Active rear spoiler.....	11
4.1.1.	System components.....	12
4.1.2.	System wiring diagram.....	16
4.1.3.	Operation and function.....	17
4.1.4.	Notes for Service.....	19
5.	Exterior Equipment.....	21
5.1.	Aerodynamics.....	21
5.2.	Panorama glass roof.....	22
6.	Interior Equipment.....	23
6.1.	Rear seats.....	23
7.	Luggage Compartment.....	24
7.1.	Parcel shelf.....	25
8.	Auto Start-Stop Function.....	26
8.1.	New features.....	26
8.2.	Longitudinal jerk.....	26
8.3.	Stop delay.....	27
8.4.	Start strategy.....	28
8.5.	Starter motor N20 engine.....	29
9.	Electric Wastegate Valve.....	30
10.	DSC Hydraulic Unit.....	31
11.	Bus System Overview.....	32

F34 Complete Vehicle

Contents.

12. Headlights.....	35
---------------------	----

F34 Complete Vehicle

1. Introduction

The BMW 3-Series Gran Turismo will be launched only with xDrive on all production models starting in July 2013. With its unique vehicle concept it extends the BMW 3-Series range with an independent variant.

The F34 will be offered exclusively with an 8 speed sport automatic transmission with shift paddles designated 8HP45 as standard, with no manual transmission option.

In the classical sense, Gran Turismo (Italian for "great travel") refers to more powerful vehicles which, in contrast to pure sports cars, used to feature more comfort, a larger interior and greater suitability for daily use on long journeys.

The seat positions have been raised, the main advantage of this is for all passengers to get in and out of the vehicle with ease. This seat position also provides the driver with a better view of traffic conditions.



BMW 3-Series Gran Turismo

F34 Complete Vehicle

1. Introduction

Special features:

- GT proportions with flatter, longer and Coupe-like roofline.
- Frameless doors without fixed side windows.
- Active rear spoiler integrated in the tailgate.
- Higher seat position for better view when driving.
- Comfortable entry and exit at front and back.
- More comfort in the rear passenger compartment and more loading volume thanks to a larger wheelbase (+ 110 mm / + 4.3 inches in comparison to F31) and longer vehicle length.
- Increase of the vehicle width.
- Rear seat designed for 3 passengers.
- One-piece tailgate for more capacity and loading options.
- Two-piece storage shelf, rear part with tailgate can be moved.
- Comfort fold down of rear seat backrests from the luggage compartment.

1.1. Models

The model range of the BMW 3-Series Gran Turismo includes the following variants:

Model	Engine	Power [kW (HP)]	Torque [Nm (lb-ft)]	Displacement [cm ³]	Exhaust emission standards
BMW 328i xDrive GT	N20B2000	180 (240)	350 (225)	1997	ULEV II
BMW 335i xDrive GT	N55B30M0	225 (300)	400 (300)	2979	ULEV II

1.2. Performance

Top speed for both variations of the F34 is electronically limited at 130 mph.

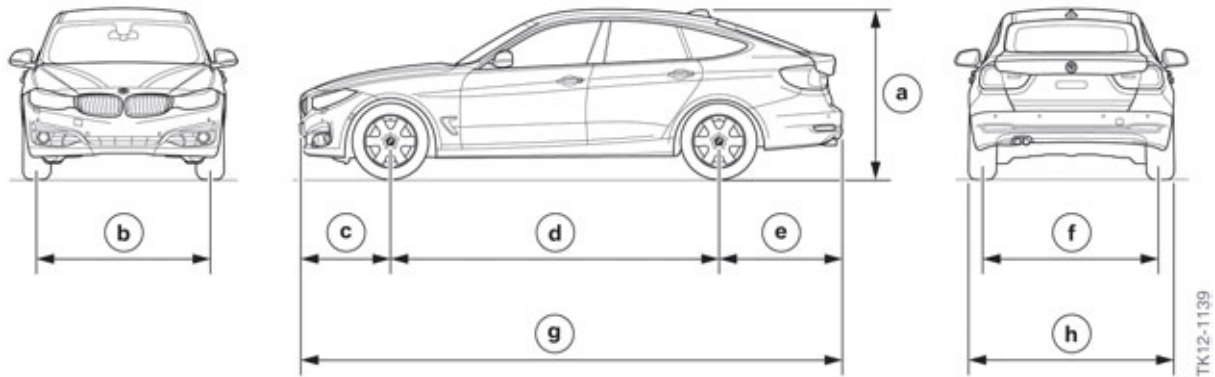
Acceleration from 0–60 mph is 6.1 seconds for the 328i xDrive and 5.1 seconds for the 535i xDrive model.

F34 Complete Vehicle

1. Introduction

1.3. Dimensions

The outer dimensions of the F34 and F31 are in the following table.



F34 outer dimensions

Index	Explanation		F34	F31
a	Vehicle height, empty	[mm]	1508	1429
b	Front track width, basic wheels	[mm]	1541	1543
c	Front overhang	[mm]	803	776
d	Wheelbase	[mm]	2920	2810
e	Rear overhang	[mm]	1101	1038
f	Rear track width, basic wheels	[mm]	1586	1583
g	Vehicle length	[mm]	4827	4624
h	Vehicle width excluding exterior mirror (vehicle width with exterior mirror)	[mm]	1828 (2047)	1811 (2031)

The F34 is longer by + 200 mm / + 7.8 inches in comparison to F31 along with the width being + 18 mm / + .7 inches wider than the F31. This allows for more comfort in the rear passenger area along with increased cargo room.

1.4. Weights and payload

Vehicle curb weights and payloads of the BMW 3-Series Gran Turismo.

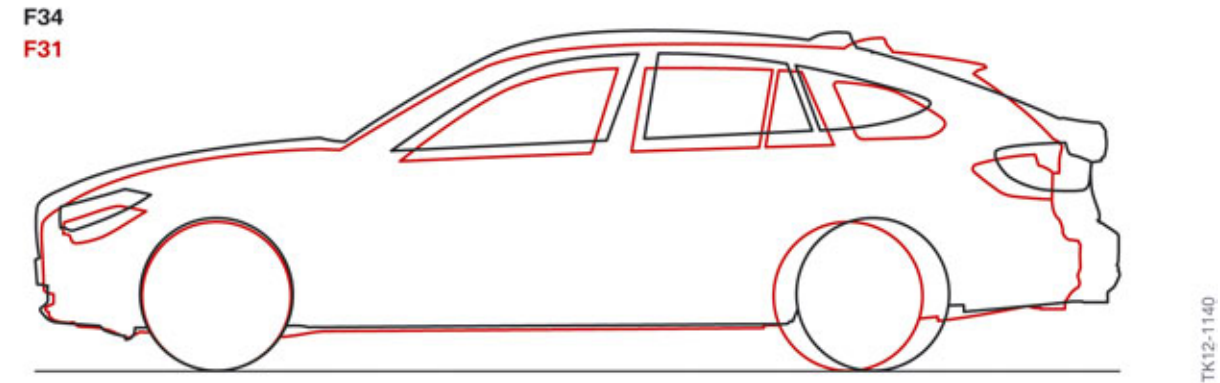
Model		Curb weight (US)	Payload
BMW 328i xDrive GT	kg (lbs)	1776 (3915)	424 (935)
BMW 335i xDrive GT	kg (lbs)	1819 (4010)	438 (965)

F34 Complete Vehicle

1. Introduction

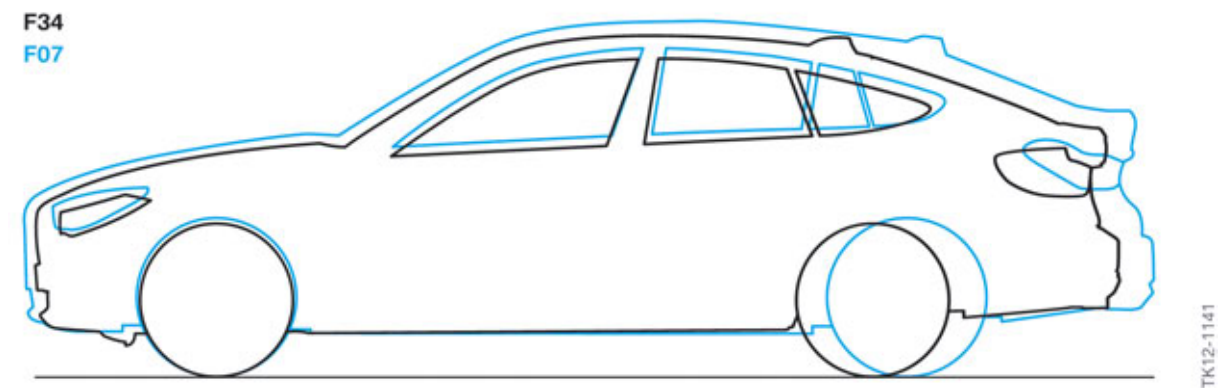
1.5. Silhouette comparison

The silhouette comparison shows that the F34 offers more space than the BMW F31 3-Series Touring, but still remains below the 5-Series Gran Turismo.



F34 Silhouette comparison with BMW F31

		F34	F31
Vehicle height, empty	[mm]	1508	1429
Front overhang	[mm]	803	776
Wheelbase	[mm]	2920	2810
Rear overhang	[mm]	1101	1038
Vehicle length	[mm]	4827	4624

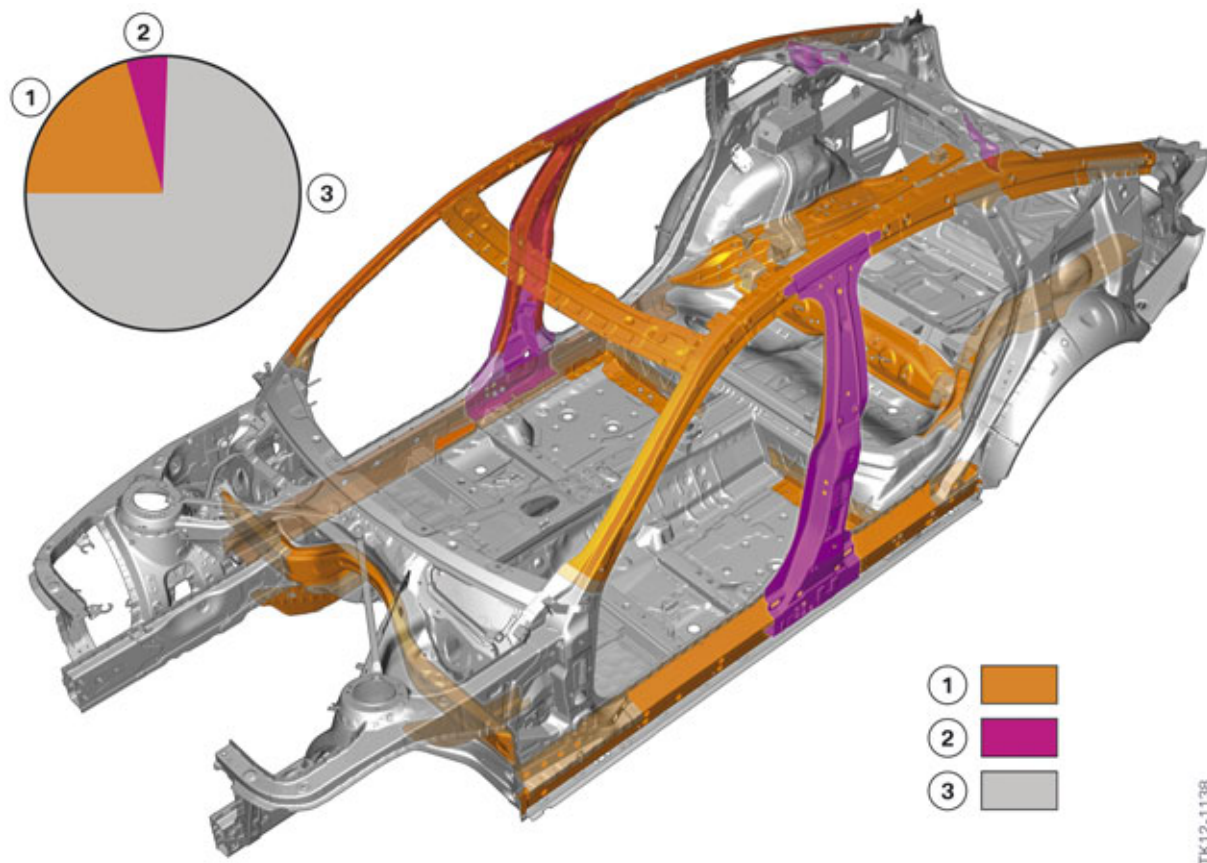


F34 Silhouette comparison with BMW F07

		F34	F07
Vehicle height, empty	[mm]	1508	1559
Front overhang	[mm]	803	838
Wheelbase	[mm]	2920	3070
Rear overhang	[mm]	1101	1090
Vehicle length	[mm]	4827	4998

F34 Complete Vehicle

2. Bodyshell



F34 material grades for bodyshell

Index	Explanation
1	Multi-phase steels (> 300 MPa), 21 % of the body weight
2	Hot-formed steels (> 900 MPa), 5 % of the body weight
3	Other steels (< 300 MPa), 74 % of the body weight

Multi-phase steels are steels where the structure consists of a number of phases. Higher-strength multi-phase steels with a yield strength: $R_{p0.2}$ of 300 to 600 MPa are, for example, dual-phase steels or TRIP steels. Super-strength multi-phase steels with a yield strength $R_{p0.2}$ in excess of 600 MPa are, for example, complex-phase steels or martensitic-phase steels. (Martensitic steel refers to a very hard form of steel crystalline structure.)

Hot-formed manganese-boron steels are super-strength steels with a yield strength $R_{p0.2}$ in excess of 900 MPa.

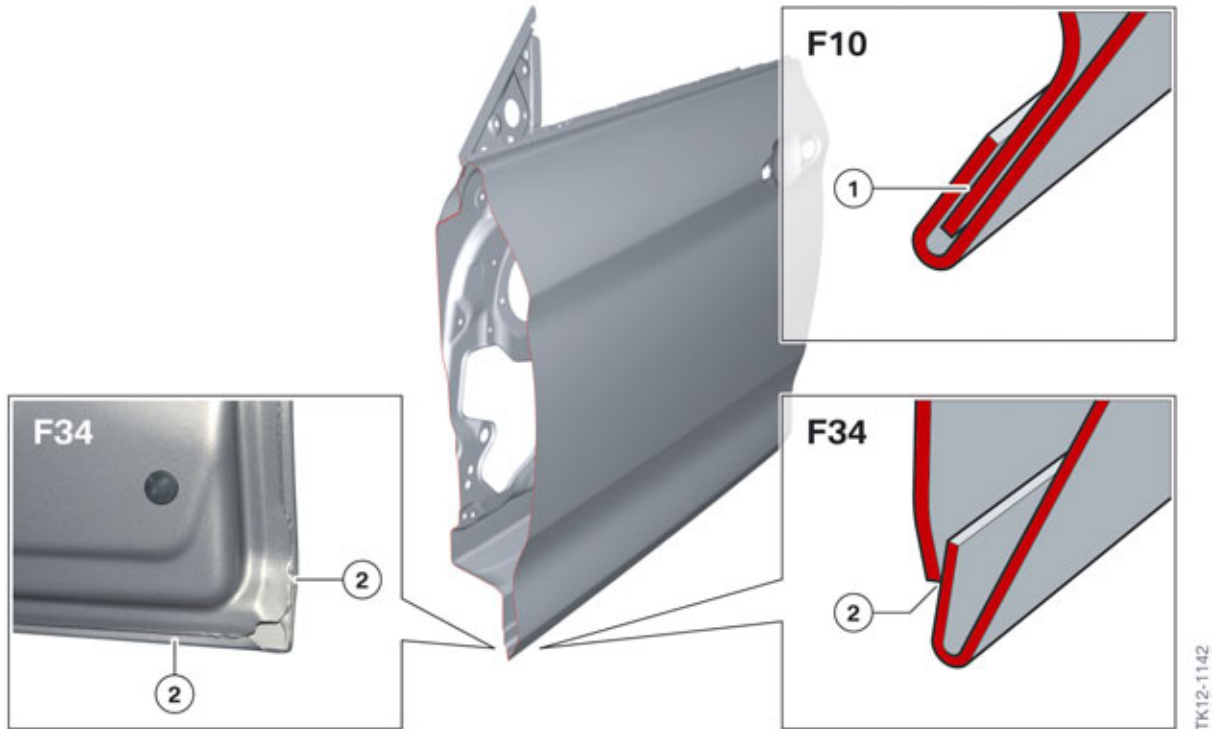
To reduce the vehicle weight and still guarantee maximum stability of the bodyshell, high-strength and super-strength multi-phase steels are used, among other things. Super-strength hot-formed steels are also used.

As a result, the weight of the bodyshell is only 353.0 kg / 776.6 lbs and the weight of the body (with doors, folds and side walls) is only 465.1 kg / 1023.2 lbs.

F34 Complete Vehicle

3. Doors

3.1. Inner door panel/Door outer skin



Assembly of F34 inner door panel/door outer skin

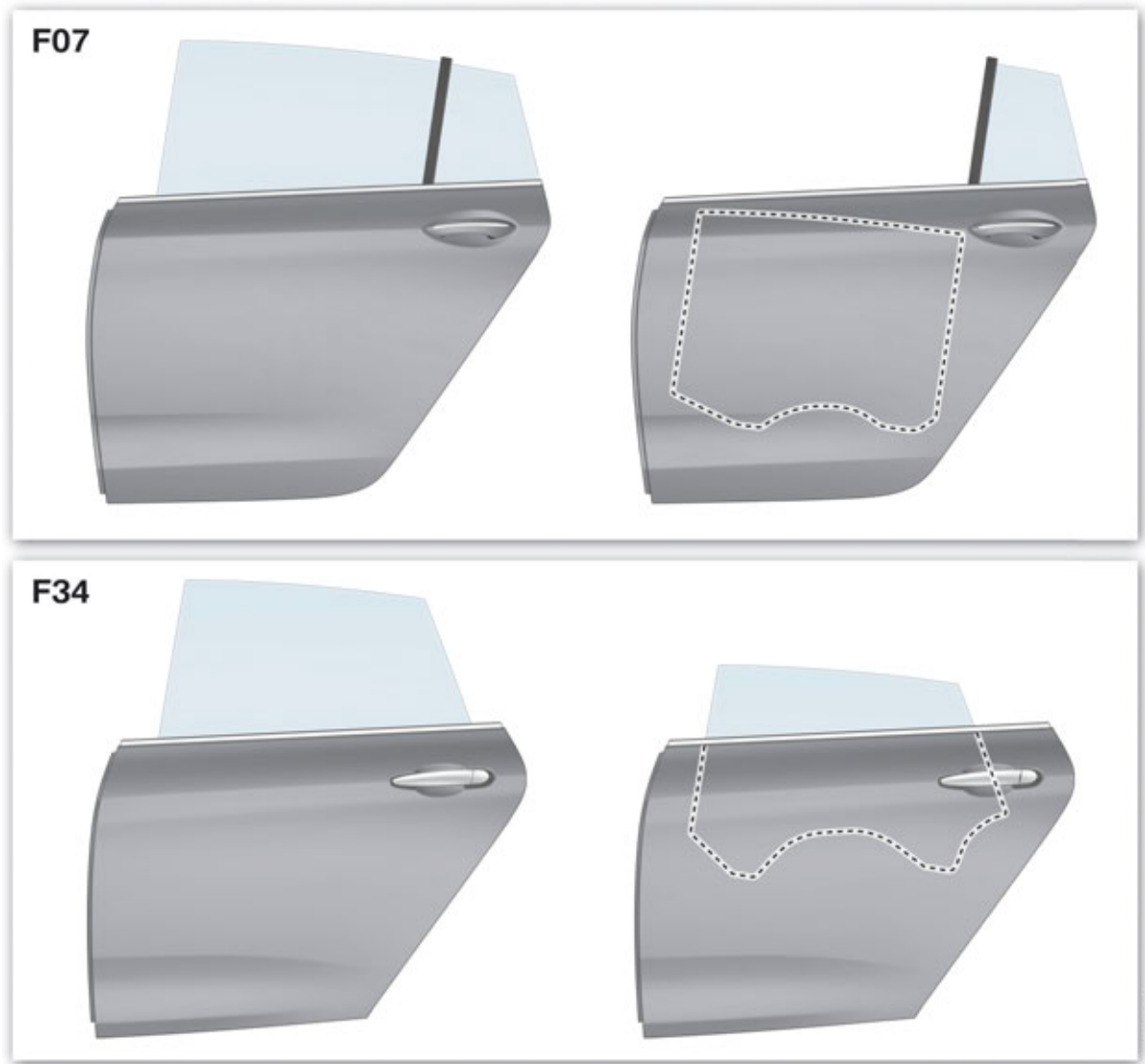
Index	Explanation
1	Folded flange with adhesive
2	Laser weld connection

For the first time in the F34, the use of flanges and adhesive in the construction of the door was changed to laser welding. BMW 3-Series Gran Turismo utilizes laser welding for the assembly of the inner door structure and door outer skin. Despite the weight reduction from laser welding, rigidity of the doors was increased and corrosion protection was also improved.

F34 Complete Vehicle

3. Doors

3.2. Side windows



F34 side windows, rear doors

The BMW 3-Series Gran Turismo is equipped with frameless doors with no fixed side windows. The side windows in the front doors can be fully lowered. Both the all-round visibility and the proportions in the side view were improved.

In comparison to the F07, the F34 side windows in the rear doors cannot be fully lowered. The rear window will extend to a maximum of 150 mm / 5.9 inches above the window cavity as seen in the picture above.

F34 Complete Vehicle

3. Doors



F34 rubber seal, side window

A rubber seal with a double sealing lip is installed for the side windows. It generates low noise when opening and closing the windows, it also has been designed to protect against water intrusion.

F34 Complete Vehicle

4. Tailgate

Similar to the F31 3-Series Touring, the F34 3-Series Gran Turismo is also equipped as standard with an automatic luggage compartment lid actuation including a Automatic Soft Close system (SCA). Here the tailgate is opened and closed by two electric spindle drives. A locking system in the control unit for the automatic luggage compartment lid actuation (HKL) ensures the tailgate is closed.

The tailgate opens automatically when it is actuated by one of the following buttons:

- Button on the remote control
- Button on the tailgate
- Button on the A-pillar

In conjunction with the optional equipment Comfort Access (option 322) the tailgate can also be opened by means of targeted foot movement. For more information refer to ST1113 F30 Complete Vehicle Reference Manual.

The tailgate closes automatically when it is actuated via the button on the inside of the tailgate.



F34 spindle drives, tailgate

The spindle drives of the automatic tailgate are housed under a plastic trim in the rear lid frame and are hardly visible to the customer.

The rear window is bonded to the steel rear lid frame.

TK12-1146

F34 Complete Vehicle

4. Tailgate



F34 rear lid opening

It is possible to significantly increase the opening area of the luggage compartment by positioning the spindle drives in the rear lid frame and by the very narrow water channel of the BMW 3-Series Gran Turismo.

Another special feature in the F34 is the extendable rear spoiler integrated in the tailgate.

F34 Complete Vehicle

4. Tailgate

4.1. Active rear spoiler



F34 active rear spoiler

BMW vehicles generally satisfy the very high requirements of driving dynamics. The special vehicle proportions of the F34 and the resulting aerodynamic influences made the use of a rear spoiler necessary.

Through the use of an active rear spoiler in the 3-Series Gran Turismo, the negative influences of the tail lift at higher speeds could be reduced by approximately 35%.

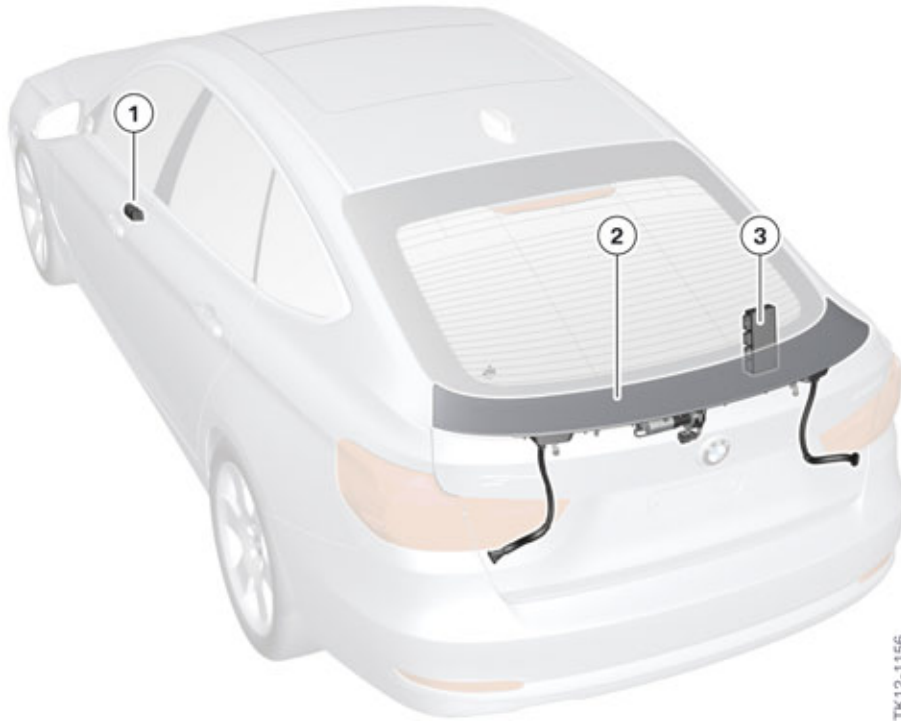
In general, lowering the tail lift leads to an increase in the drag coefficient. In the F34, however, the drag coefficient was improved slightly through intensive fine tuning of the height, contact angle and geometry of the rear spoiler.

The active rear spoiler is included in the standard equipment of the F34 and is integrated in the tailgate.

F34 Complete Vehicle

4. Tailgate

4.1.1. System components

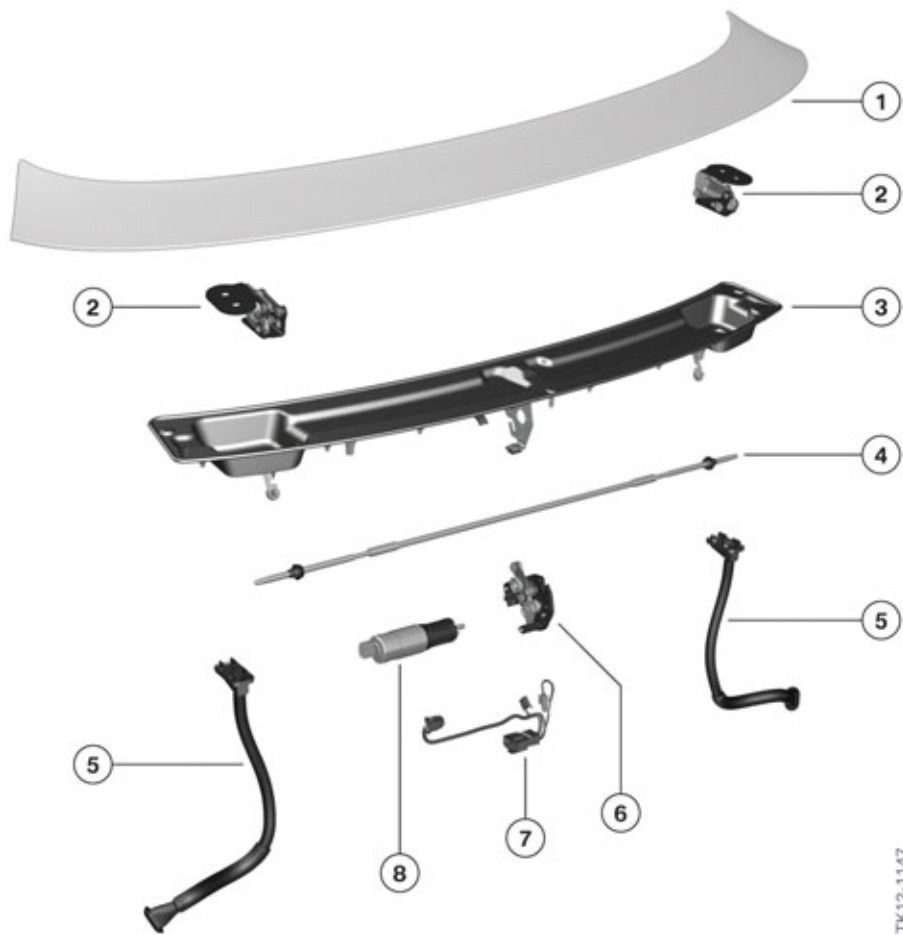


F34 components of the active rear spoiler

Index	Explanation
1	Switch block, driver's door
2	Spoiler module
3	Control unit for automatic operation of tailgate (HKL)

F34 Complete Vehicle

4. Tailgate



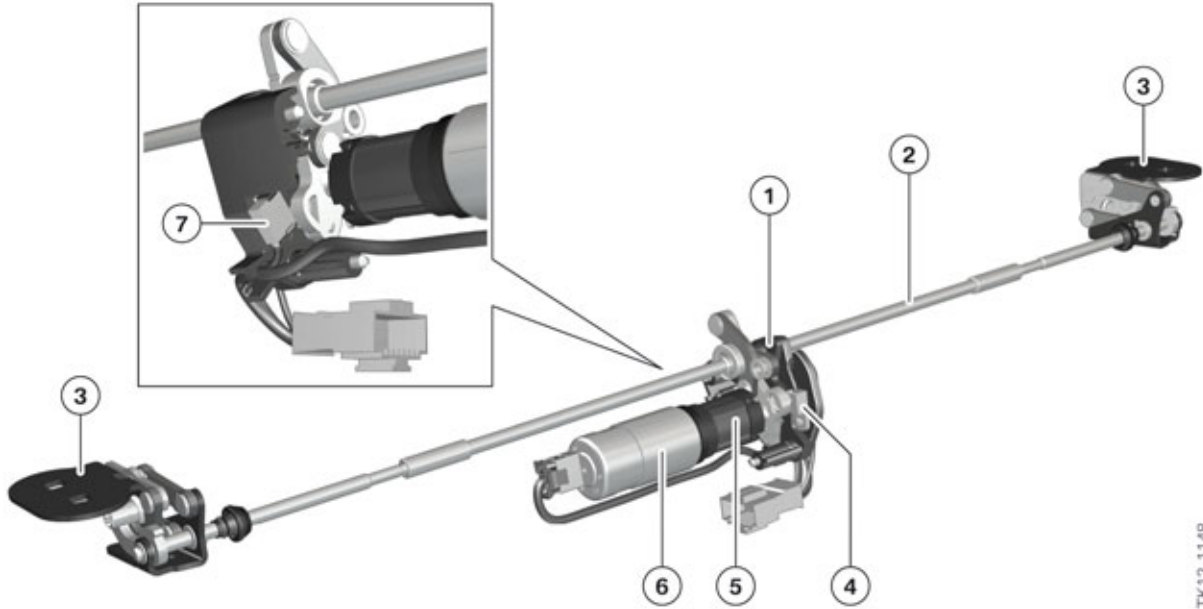
F34 components of the spoiler module

Index	Explanation
1	Spoiler blade
2	Four-way linkage hinge with spoiler blade connection
3	Carrier plate with engine connection, fastening and positioning bolts
4	Drive shaft
5	Water channel
6	Kinematics
7	Electrical lines and sensors
8	Electric motor with transmission

F34 Complete Vehicle

4. Tailgate

Kinematics and sensor system



F34 kinematics and sensor system of active rear spoiler

Index	Explanation
1	Kinematics
2	Drive shaft
3	Four-way linkage hinge with spoiler blade connection
4	Hall effect sensor (rear spoiler raised)
5	Transmission
6	Electric motor with hall effect sensor (motor rotations)
7	Microswitch (rear spoiler lowered)

The active rear spoiler is driven by an electric motor. It drives the drive shafts via a transmission and kinematics. They in turn actuate the two four-way linkage hinges, to which the spoiler blade is secured.

A hall effect sensor detects the rotations in the electric motor. Another hall effect sensor at the kinematics detects the raised end position of the rear spoiler. They are proximity-type sensors and are insensitive to temperature fluctuations, oscillations and vibrations.

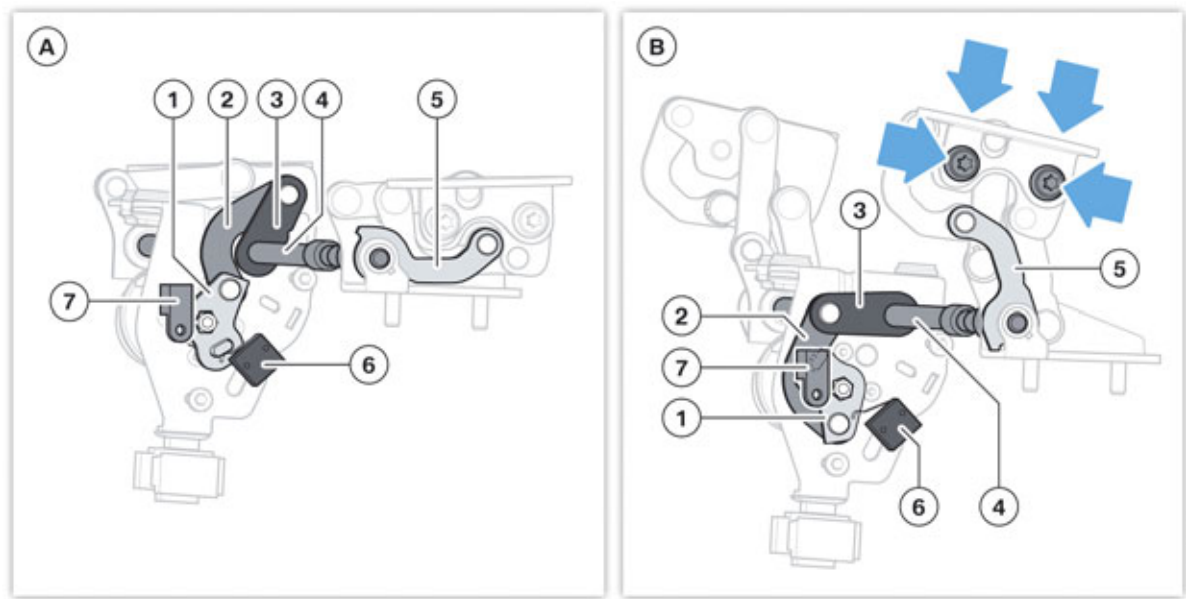
A microswitch, also at the kinematics, detects the lowered end position of the rear spoiler via a mechanical contact.

The sensors report the respective end positions and the rotations to the control unit for the automatic luggage compartment lid actuation (HKL).

The control unit for automatic luggage compartment lid actuation (HKL) controls and monitors all electrical functions of the active rear spoiler.

F34 Complete Vehicle

4. Tailgate



F34 kinematics of the active rear spoiler

Index	Explanation
A	Rear spoiler lowered (down)
B	Rear spoiler raised (up)
1	Drive arm
2	Relay lever
3	Lever
4	Drive shaft
5	Lever (four-way linkage hinge)
6	Microswitch (rear spoiler lowered/down)
7	Hall effect sensor (rear spoiler raised/up)

When the rear spoiler is being raised the transmission-output shaft rotates the drive arm (1), which then actuates the lever (3) at the drive shaft (4) via the relay lever (2). The drive shaft then actuates the lever (5) of the two four-way linkage hinges, which ultimately lift the spoiler blade.

The position of the spoiler blade is predetermined by the position and length of the lever in the four-way linkage hinge. Fine adjustment of the spoiler blade can be adjusted using the screws at the two four-way linkage hinges marked in the graphic.

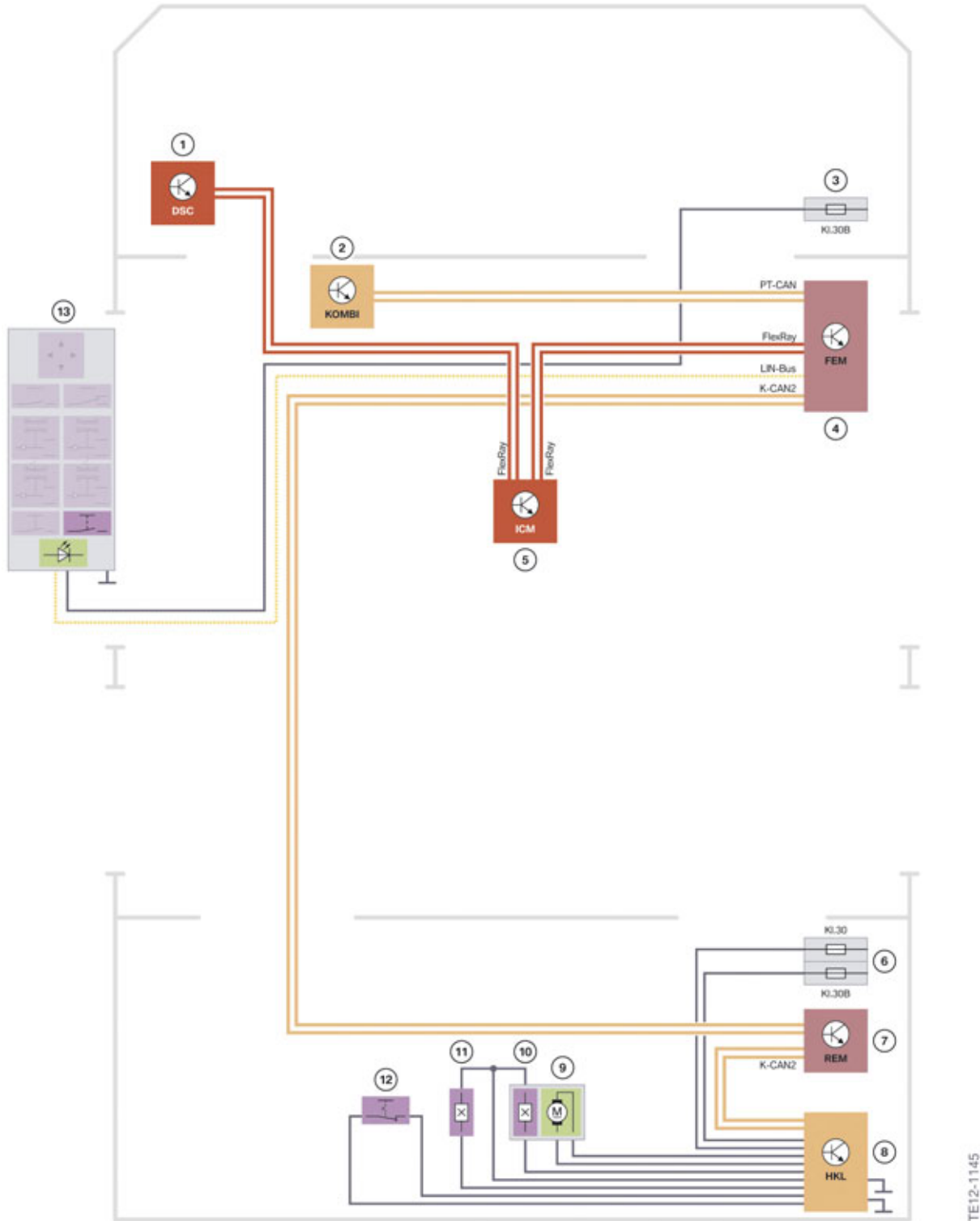
When the rear spoiler is lowered a self-locking mechanism (prevents rotation) is generated by the special shape of the relay lever (2), making intrusion from the outside impossible and guaranteeing thief proofing.

When the rear spoiler is raised a self-locking mechanism is also generated by the design of the four-way linkage hinge. As a result, an undesired retraction of the rear spoiler is prevented, also in the event of a malfunction of other components.

F34 Complete Vehicle

4. Tailgate

4.1.2. System wiring diagram



F34 system wiring diagram of active rear spoiler

F34 Complete Vehicle

4. Tailgate

Index	Explanation
1	Dynamic Stability Control (DSC)
2	Instrument cluster (KOMBI)
3	Power distribution box, engine compartment
4	Front Electronic Module (FEM)
5	Integrated Chassis Management (ICM)
6	Power distribution box, luggage compartment
7	Rear Electronic Module (REM)
8	Automatic luggage compartment lid actuation (HKL)
9	Electric motor
10	Hall effect sensor (motor rotations)
11	Hall effect sensor (rear spoiler raised/up)
12	Microswitch (rear spoiler lowered/down)
13	Switch block, driver's door

4.1.3. Operation and function

The rear spoiler is always fully lowered or raised during normal operation, intermediate settings are not intended. The activation can be controlled automatically or manually. The manual mode can be used, for instance, to clean the rear spoiler from terminal R or when the engine is running.

- **Automatic mode**
 - Driving speed = 110 km/h / 68 mph: Rear spoiler raised.
 - Driving speed < 70 km/h / 43 mph: Rear spoiler lowered.
- **Manual mode**
 - Displayed by the indicator light in the rear spoiler button.
 - Rear spoiler is always raised.

The system remains in the last mode during a terminal change, also upon a vehicle restart.

F34 Complete Vehicle

4. Tailgate



F34 Switch block, driver's door

Index	Explanation
1	Rear spoiler button

Switch between automatic and manual mode by touching the rear spoiler button in the switch block at the driver's door:

Switching from automatic to manual mode

- Touch the rear spoiler button
 - The system switches to Manual mode.
 - The indicator light in the rear spoiler button lights up.
 - The rear spoiler raises when stationary or at a driving speed < 110 km/h / 68 mph.

Switching from manual to automatic mode

At a driving speed of less than 20 km/h / 12 mph it is necessary to press and hold button until the rear spoiler is completely lowered, this is to avoid trapping (e.g. a hand) when the vehicle is rolling.

- **Driving speed ≥ 20 km/h / 12 mph:** Touch rear spoiler button
 - The system switches to Automatic mode.
 - The indicator light in the rear spoiler button goes out.
 - At a driving speed ≤ 70 km/h / 43 mph the rear spoiler is lowered.
- **Driving speed < 20 km/h / 12 mph:** Press and hold down the rear spoiler button
 - The system switches to Automatic mode.
 - The rear spoiler is lowered as long as the rear spoiler button is pressed (if the rear spoiler button is released before the rear spoiler is fully lowered, the rear spoiler is fully raised again).
 - If the rear spoiler is fully lowered, the indicator light in the rear spoiler button goes out.

F34 Complete Vehicle

4. Tailgate

In vehicles with Comfort Access (option 322) the rear spoiler can be lowered when a vehicle is stopped by using the convenient closing function. After the detection of your finger on the door handle the windows will close if previously open and the spoiler will lower.

If the convenient closing function is interrupted by removing your finger from the contact area, the rear spoiler is fully raised again, if the rear spoiler was not fully lowered.

4.1.4. Notes for Service

Speed limits

Speed	Designation	Comment
3 km/h (1.8 mph)	Convenient closing	Convenient closing is only possible below this speed limit.
20 km/h (12 mph)	Constant pressure	It is necessary to press the button constantly for manual lowering below this speed limit. The speed limit is also used for automatic lowering of the rear spoiler after a fault.
50 km/h (31 mph)	Self-test/ production, transport and workshop mode	Speed limit for the self-test after engine start. The rear spoiler is activated briefly (not visible). Below this speed limit or if a driving speed is not detected, the rear spoiler neither lowers or raises in the production, transport and workshop mode.
70 km/h (43 mph)	Retract rear spoiler	At or below this speed limit the rear spoiler lowers again in Automatic mode.
110 km/h (68 mph)	Extend rear spoiler	The rear spoiler raises at this speed limit.
130 km/h (80 mph)	Continuous gong	A continuous gong sounds for 90 seconds in the event of a spoiler fault at this speed limit.

Check Control messages

Malfunctions at the rear spoiler are displayed by a Check Control message and acoustic signals (also when production, transport and workshop mode is set):

Check Control symbol	Check control message	Text in central information display
	Rear spoiler	Rear spoiler faulty. Do not drive faster than 130 km/h (80 mph). Possible causes, see Owner's Handbook.

In addition to the Check Control message shown above, an information gong sounds once. A continuous gong sounds for 90 seconds from a driving speed of 130 km/h / 80 mph.

F34 Complete Vehicle

4. Tailgate

A malfunction is detected in the event of the following faults:

- 5 seconds after the speed limit is exceeded "Raise rear spoiler" not detected "Rear spoiler is raised".
- Interruption of the signal "Rear spoiler raised", despite exceeded speed limit "Raise rear spoiler".
- Supply voltage of control unit HKL < 9 V when engine is running and exceeded driving speed "Raise rear spoiler".
- Speed signal faulty.
- HKL control unit faulty (the HKL control unit detects its own fault and sends the Check Control message).
- HKL control unit no longer receives any CAN messages (e.g. the HKL control unit can still send Check Control messages).
- Sensor fault "Rear spoiler is raised".
- HKL control unit not encoded.
- Rear spoiler cannot be raised manually (e.g. frozen).
- Rear spoiler in an intermediate position setting, when attempting to retract (once for each terminal connection cycle, terminal R).
- HKL control unit has set the production, transport and workshop mode for the rear spoiler.

From a driving speed of 50 km/h / 31 mph the DSC is switched back on automatically if previously turned off. This is for safety reasons in the event of a malfunction with the rear spoiler. Vehicle stability can be compromised at speeds greater than 50 km/h / 31 mph without the function of the rear spoiler.

Service position of the rear spoiler

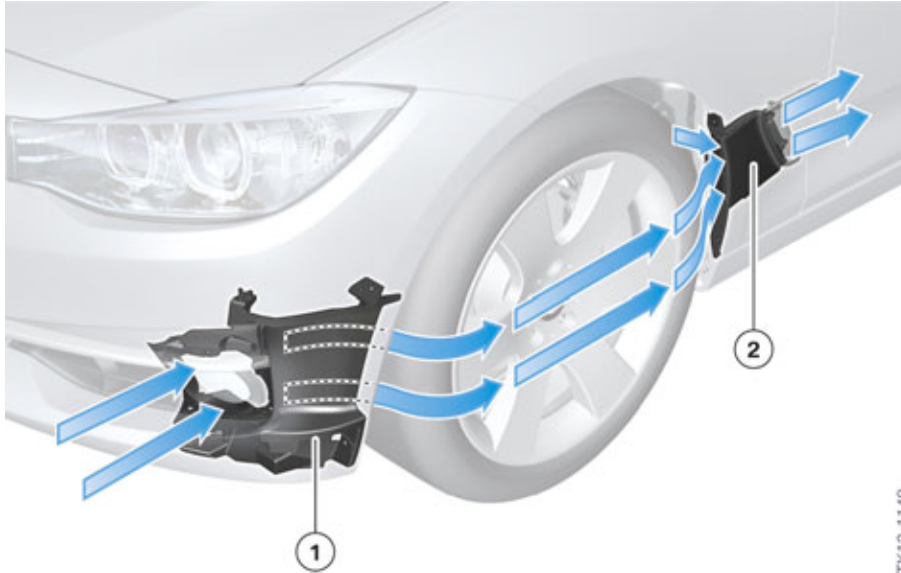
To facilitate the removal and installation of the rear window, the rear spoiler can be brought into a service position via the BMW workshop system ISTA.

In the service position the rear spoiler is in "Manual mode, intermediate setting", so that it does not move when the vehicle is stationary. The service position can be removed again by lowering the rear spoiler using the rear spoiler button or the rear spoiler test module in ISTA.

F34 Complete Vehicle

5. Exterior Equipment

5.1. Aerodynamics



F34 air, aerodynamics

Index	Explanation
1	Air duct ("Air Curtain")
2	Air duct ("Air Breather")

For the first time at BMW an air duct has been installed behind the front wheel in the fender. The "Air Breather" reduces the inflow in the rear area of the wheel arch and reduces air turbulence in the wheel arch area. As a result, the drag of the vehicle is reduced, as well as the fuel consumption, and also lowers CO₂ emissions.

The high amount of air flow which enters the front wheel arch, are guided through the air duct to the fender via a vertical air inlet. At corresponding pressure ratios there is an advantage of the drag coefficient. The higher the pressure there is, the more effective the "Air Breather".

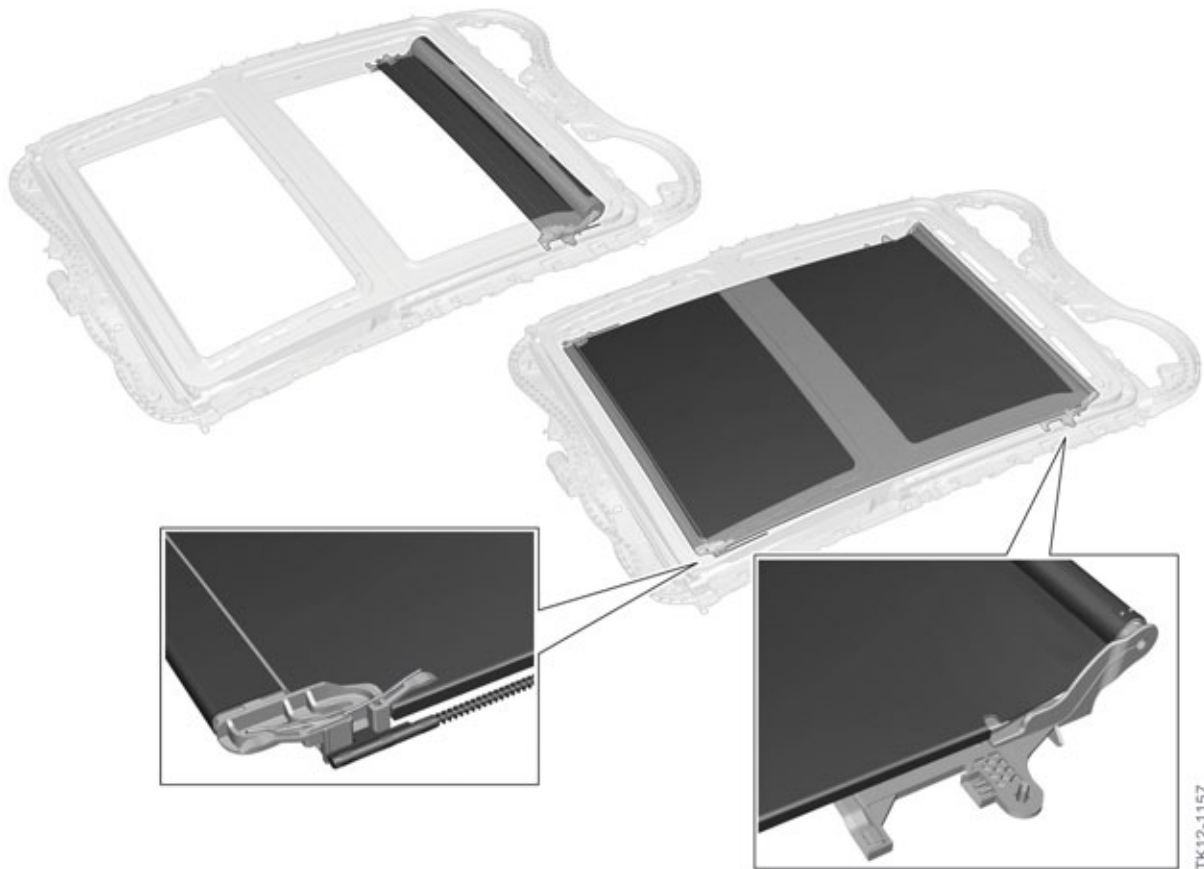
The "Air Breather" is very effective when combined with the "Air Curtain".

F34 Complete Vehicle

5. Exterior Equipment

5.2. Panorama glass roof

In comparison to a traditional sliding roof system, the panorama glass roof makes a positive contribution to the interior of the vehicle through an enlargement of the glass area. More headroom is also created by using a roller type sunblind. The roller sunblind is installed in the headlining cover and thus has a high-quality effect on the interior design.



F34 roller, panorama glass roof

The material of the roller sunblind is made from polyethylene. The inside layer that is exposed to the interior corresponds to the interior color of the headliner, the outer layer that is exposed to the outside is black. The roller sunblind not only blocks out the sun, it also reduces interior noise when fully closed. When the glass slide/tilt sunroof panel and roller sunblind are closed fully, the noise level is similar to that of a panorama glass roof with a conventional sliding head liner.

To rule out the risk of possible injuries, an anti-trap mechanism was integrated for the glass slide/tilt sunroof panel and the roller sunblind in the electronics.

F34 Complete Vehicle

6. Interior Equipment

6.1. Rear seats



F34 rear seats

The larger vehicle width in comparison to the F30 and F31 improved the seating capacity, allowing 3-person occupancy of the rear seat. The seat belt buckles are more accessible and the centre rear seat backrest can now be folded down when child seats are installed with the ISOFIX child restraint system.

The rear seat backrest in the F34 can be divided in the ratio 40/20/40. The backrest angle can also be adjusted in 15 positions, this allows the luggage compartment to be increased by up to 33 liters. Unlocking the rear seat backrests from the luggage compartment is possible. The backrests are also dampened by a shock absorber.



F34 unlocking, rear seat backrest

F34 Complete Vehicle

7. Luggage Compartment

Special features

- Luggage compartment capacity 520 – 1600 liters (+ 25 liters in comparison to the F31).
- Luggage compartment floor with smooth transitions to the side trim panels.
- Luggage compartment floor with flush transition to the folded rear seat backrest for the transport of larger items.
- Large loading hatch.
- Two-piece storage shelf, can be stowed underneath the luggage compartment floor.
- Side flap on the left for access to the tool kit.
- Multifunction tray below the floor panel.
- Floor panel supported by gas pressure spring, in vehicles with storage compartment package (option 493).
- Comfortable access to the fuses via a separate lid in the multifunction tray.
- LED light bands in the trim panels of shoulder, for vehicles with light package (option 563).
- Multifunction hooks, lashing eyes and power socket (12 V).



F34 LED light bands, luggage compartment

F34 Complete Vehicle

7. Luggage Compartment

7.1. Parcel shelf



F34 storage shelf

Index	Explanation
1	Parcel shelf, rear part
2	Parcel shelf, front section

The parcel shelf separates the passenger compartment from the luggage compartment. In the BMW 3-Series Gran Turismo a two-piece parcel shelf is installed. The front part is secured on both sides using spring clips at the trim panel. The rear part is attached to the tailgate and is attached by two detent pins.

Both parcel shelves can be removed and stowed under the luggage compartment floor to increase the luggage compartment capacity.

F34 Complete Vehicle

8. Auto Start-Stop Function

The BMW 3-Series Gran Turismo is equipped with the automatic engine start-stop function. The operating logic is known from current BMW models.

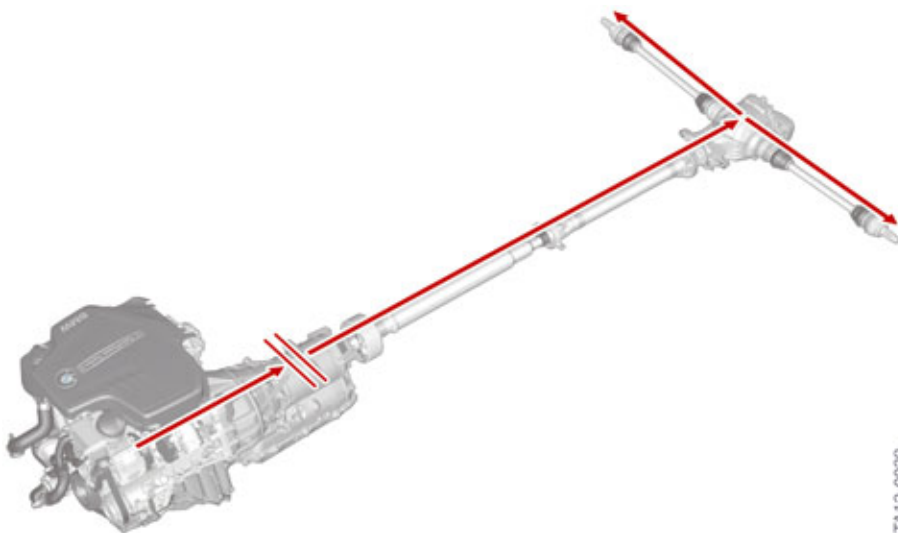
8.1. New features

Two new features have been added to the Auto Start-Stop Function. These features will also be added to all current BMW models before the series launch of the F34.

They are referred to as:

- Longitudinal Jerk
- Stop Delay

8.2. Longitudinal jerk



BMW drive train

The drive train is pre-loaded by the engine torque and the stationary wheels, as soon as a drive position (R, D, S or M) is selected when the engine is running and the brake pedal is pressed. If the engine is stopped by the automatic engine start-stop function in this state, the mechanical transmission oil pump is also no longer powered. As not enough transmission oil pressure can build up below a certain engine speed, a multidisc clutch in the automatic transmission is opened very quickly. This in turn means that the torque at the drive train drops suddenly, whereby a longitudinal jerk of the vehicle is felt.

To avoid this longitudinal jerk, a multidisc clutch is slowly opened in the automatic transmission immediately before the automatic engine shutdown which improves the comfort significantly.

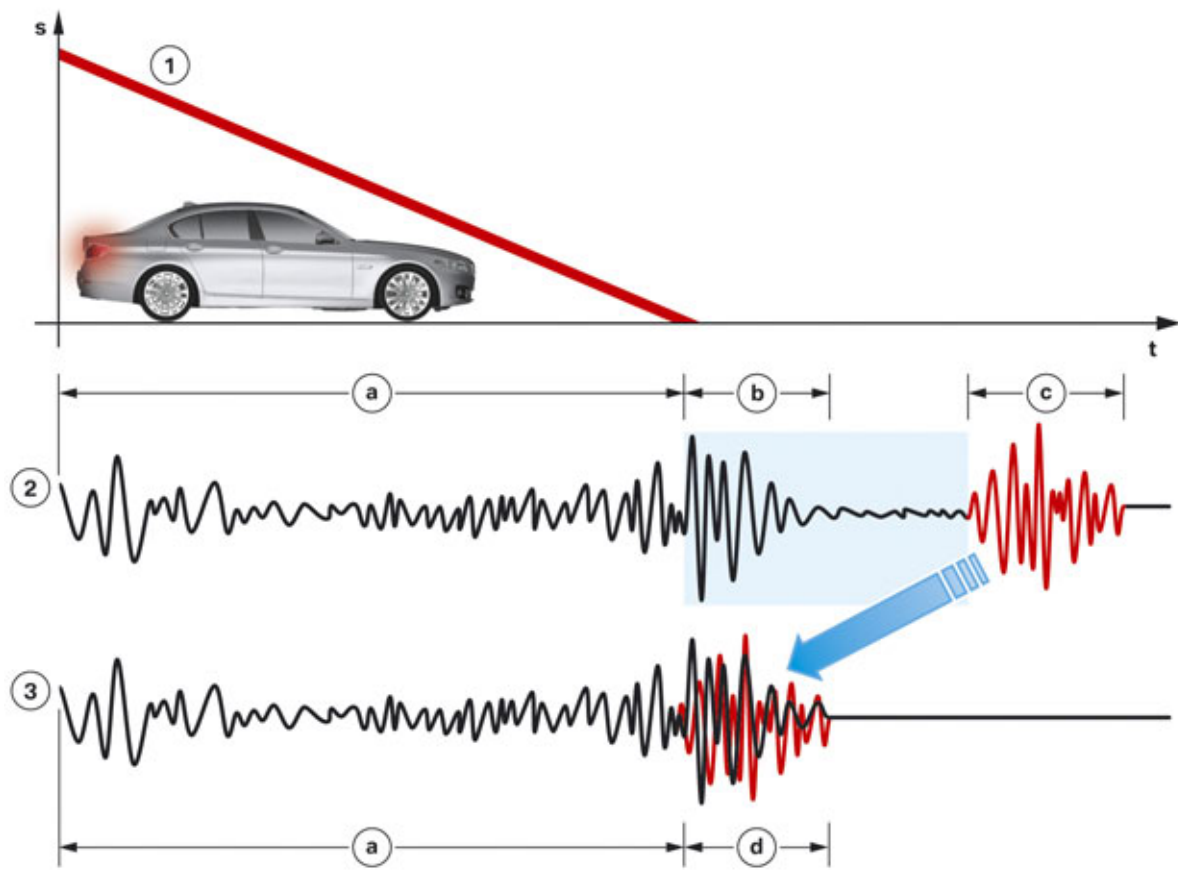
This additional function was introduced in all vehicles with an automatic engine start-stop function and automatic transmission around July 2013.

F34 Complete Vehicle

8. Auto Start-Stop Function

Other changes to the automatic engine start-stop function are introduced parallel to the model revision. In general, the time delay between vehicle standstill and an automatic engine shutdown is deleted. In this case the multidisc clutch in the automatic transmission is not opened before the automatic engine shutdown.

8.3. Stop delay



Automatic engine start-stop function without stop delay

Index	Explanation
1	Vehicle speed
2	Vehicle excitation (automatic engine start-stop function with stop delay)
3	Vehicle excitation (automatic engine start-stop function without stop delay)
a	Roadway excitation
b	Stopping jerk of the vehicle
c	Stopping the engine
d	Stopping jerk of the vehicle and engine (masked)

F34 Complete Vehicle

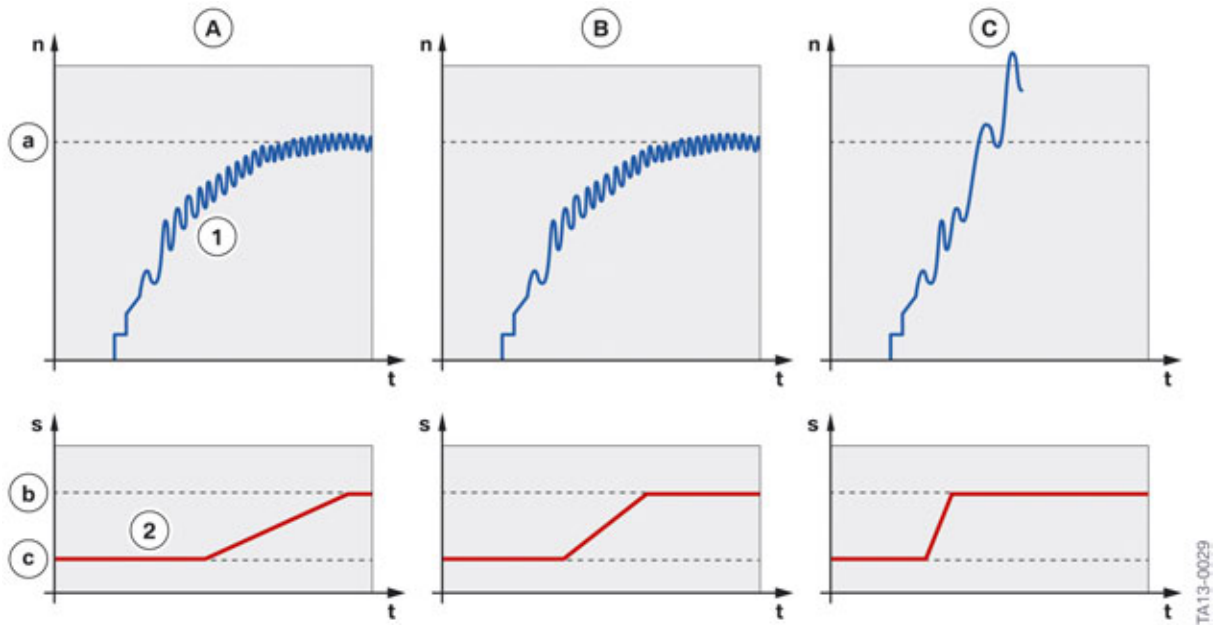
8. Auto Start-Stop Function

Similar to vehicles with manual gearbox, the time delay of approximately one second until the automatic engine shutdown (light blue area in the graphic) is now also deleted in vehicles with an automatic transmission. The engine is shut down immediately upon vehicle standstill ($v = 0$ mph). The shutdown of the engine is hardly noticeable as it is masked by the stopping jerk of the vehicle.

In special cases, the automatic engine shutdown can still be effected shortly after the vehicle standstill, for example:

- by the driver such as steering wheel movement (for maintaining the power steering support)
- upon heavy braking (from 5 m/s^2)
- on inclines or downhill gradients (from approx. 3.5 %).

8.4. Start strategy



Automatic engine start-stop function start strategy

Index	Explanation
A	System start
B	Convenient starting
C	Dynamic start
1	Engine speed
2	Position of multidisc clutch
a	Idle speed
b	Multi-disc clutch closed
c	Multi-disc clutch open

F34 Complete Vehicle

8. Auto Start-Stop Function

A new start strategy for the automatic engine start-stop function will be introduced approximately July 2013. The automatic engine start is optimized depending on the situation. Depending on the prerequisites, more emphasis is placed on comfort and quick drive off.

System start	Convenient start	Dynamic start
The automatic engine start is effected by a system switch-on request (e.g. by air-conditioning), the brake pedal remains pressed.	The automatic engine start is effected by releasing the brake, the accelerator pedal is not pressed.	The automatic engine start is effected by releasing the brake, the accelerator pedal is pressed for drive off.
The engine speed is slowly increased until it reaches the idle speed.	The engine speed is slowly increased until it reaches the idle speed.	The engine speed is increased quickly.
The multidisc clutch in the automatic transmission closes very slowly.	The multidisc clutch in the automatic transmission closes slowly.	The multidisc clutch in the automatic transmission closes quickly.
A longitudinal jerk noticeable to the driver is avoided.	Smooth and comfortable drive-off is made possible.	Quick drive off is therefore made possible.

8.5. Starter motor N20 engine

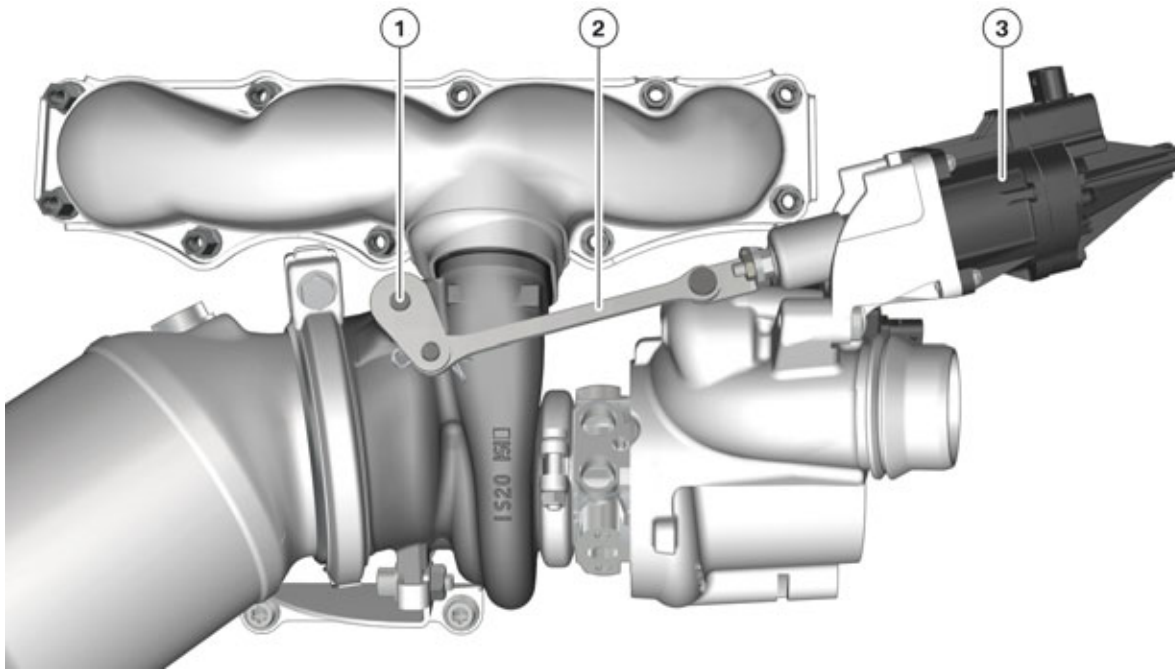
In vehicles with an N20 engine a new, more powerful starter motor is used, which enables the new starter motor to be actuated before the engine standstill, at an engine speed of 0 to approximately 150 rpm.

This is primarily used with a recognized switch-on prompt during an automatic engine shutdown by the automatic engine start-stop function. The customer therefore does not have to wait if he wants to continue driving straight away after an automatic engine shutdown, (e.g. at a stop sign).

Further information on the automatic engine start-stop function is available in the Technical Training Reference Manual ST1301 F31 Complete Vehicle.

F34 Complete Vehicle

9. Electric Wastegate Valve



F34 electrical wastegate valve

TA12-1161

Index	Explanation
1	Wastegate valve
2	Linkage rod
3	Wastegate electric actuator

The wastegate valve prevents excessive charging and turbine pressure at high engine speeds. To comply with the exhaust emission standards an electrical wastegate valve is being used in the N20 ULEV II and N26 SULEV engine as of 7/2012, the N55 engine will also get the electrical wastegate starting in 7/2013 in the following models F30, F31, F30H, F34. Other BMW models with the N55 engine will also be using the electric wastegate in the near future.

The DME controls and monitors the actuator of the wastegate valve. It contains a direct current motor for adjusting the linkage and a potentiometer for determining the traverse path. The vacuum reservoir in the engine acoustic cover has been eliminated.

The wastegate valve can be controlled quicker and more precisely with the electrical activation. It is no longer exposed to the influences of the air pressure fluctuations and can be diagnosed better, (e.g. by detection of the flap angle for fresh air and the power consumption).

The catalytic converter heating period was reduced by the larger opening angle of the wastegate valve. This is also supported by the shortening of the turbine housing, whereby the catalytic converter could be positioned closer to the wastegate valve.

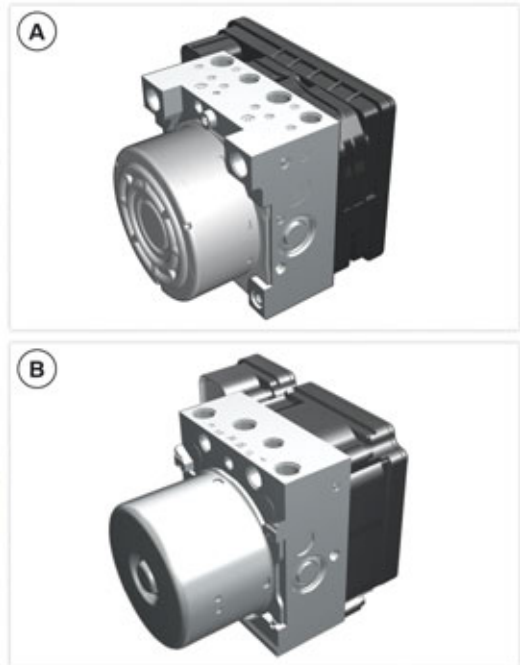
Further information on the electric wastegate function is available in the Technical Training Reference Manual ST1215 N26 Engine.

F34 Complete Vehicle

10. DSC Hydraulic Unit



F34 DSC hydraulic unit



TF12-1160

Index	Explanation
A	DSC hydraulic unit MK100
B	DSC hydraulic unit MK60

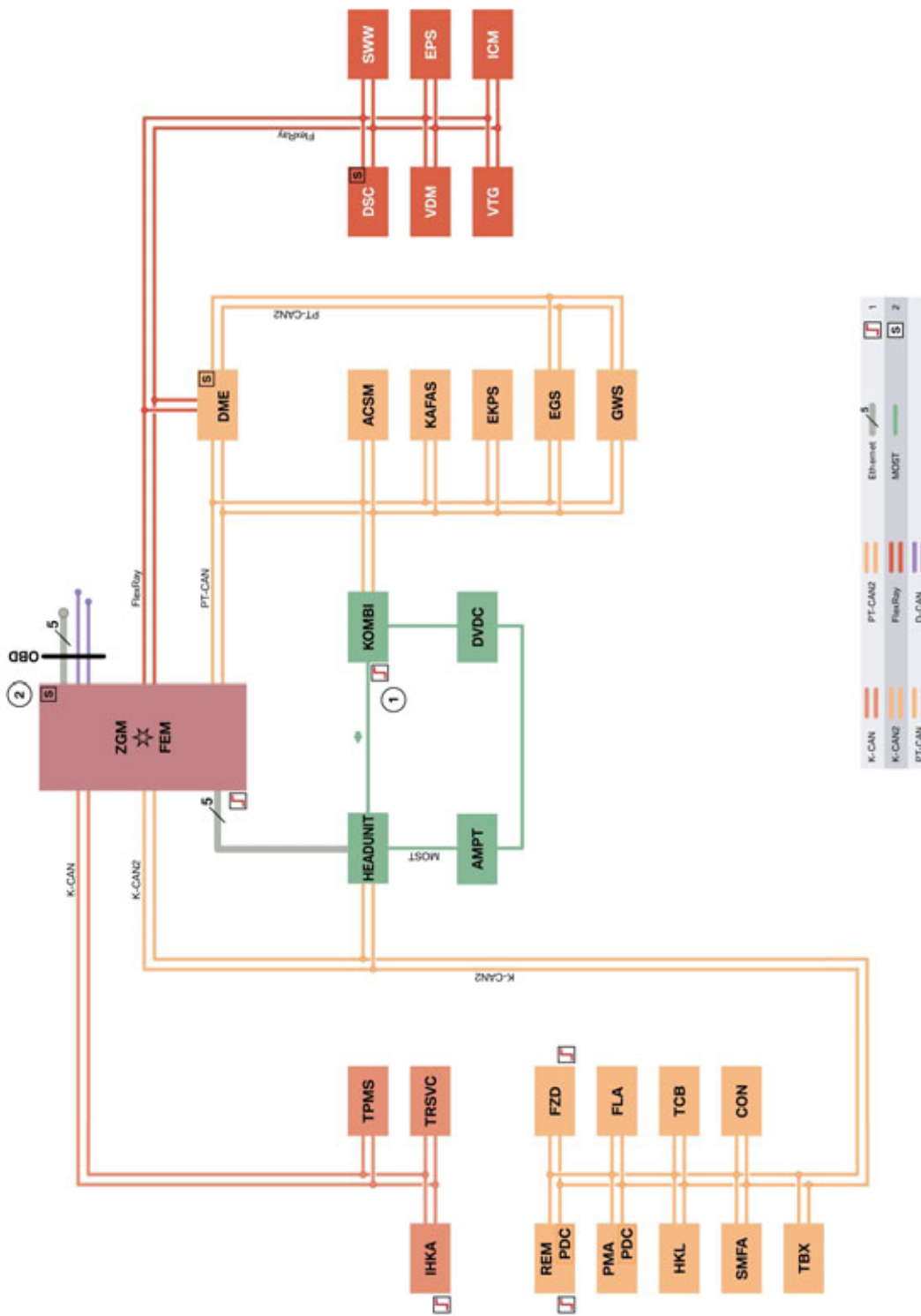
In the F34 the new compact DSC hydraulic unit MK100 is used. It is gradually replacing the hydraulic unit MK60 in other vehicles. In addition to the lower manufacturing costs, the hydraulic unit MK100 has a lower weight (by up to 500 g / 1.1 lbs depending on the variant) in comparison to the hydraulic unit MK60 and requires less installation space.

In comparison to the variant MK60A with active cruise control (ACC), two of the three pressure sensors installed could be deleted. Depending on the variant, only one pressure sensor still exists in the new DSC hydraulic unit.

With regard to Service there are no changes in relation to the maintenance or repair of the system. Only the position, the attachment points of the brake lines and the connector connection at the DSC control unit were changed. Similar to the predecessor, the MK100 DSC control unit is attached to the hydraulic unit, which can be replaced individually.

F34 Complete Vehicle

11. Bus System Overview



TE12-1159

F34 bus overview

F34 Complete Vehicle

11. Bus System Overview

Index	Explanation
1	Control units with wake-up authorisation
2	Start-up node control units for starting and synchronizing the FlexRay bus system
ACSM	Advanced Crash Safety Module (ACSM)
AMPT	Top HiFi amplifier
CON	Controller
D-CAN	Diagnosis-on-Controller Area Network
DME	Digital Motor Electronics
DSC	Dynamic Stability Control
DVDC	DVD changer
EGS	Electronic transmission control
EKPS	Electronic fuel pump control
EPS	Electromechanical Power Steering
Ethernet	Cable-based data network technology for local data networks
FEM	Front Electronic Module
FLA	High-beam assistant
FlexRay	Fast, preset and fault-tolerant bus system for use in automotive sector
FZD	Roof function center
GWS	Gear selector
HEADUNIT	Headunit (Headunit Basic or Headunit High)
HKL	Automatic luggage compartment lid actuation
ICM	Integrated Chassis Management
IHKA	IHKA
K-CAN	Body controller area network
K-CAN2	Body controller area network 2
KAFAS	Camera-based driver support systems
KOMBI	Instrument cluster
MOST	Media Oriented System Transport
OBD	On-board diagnosis (diagnostic socket)
PDC	Park Distance Control (with option 5DP, Parking Manoeuvring Assistant: integrated in the Parking Manoeuvring Assistant control unit, otherwise integrated in the Rear Electronic Module control unit)
PMA	Parking Manoeuvring Assistant
PT-CAN	Powertrain controller area network
PT-CAN2	Powertrain controller area network 2
REM	Rear Electronic Module

F34 Complete Vehicle

11. Bus System Overview

Index	Explanation
SMFA	Seat module, driver
SWW	Lane change warning
TBX	Touchbox
TCB	Telematic Communication Box
TPMS	Tire pressure control
TRSVc	Control unit for all-round vision cameras
VDM	Vertical Dynamics Management
VTG	Transfer box control module
ZGM	Central gateway module

In all 3-Series vehicles and other BMW models with the option 609 (Navigation System) the new Headunit High “HU-H” will be installed from approximately July 2013. Together with the Headunit High a controller with a touchpad will be available at start of production as well as an additional control unit “Touchbox TBX” for the evaluation of the inputs.

Further information on the Headunit High and the controller with touchpad can be found in the Technical Training Reference Manual ST1211 Headunit High.

The control units CON and driver's seat module SMFA will also be connected to the K-CAN2.

F34 Complete Vehicle

12. Headlights

The F34 is equipped with halogen headlights as standard. The bi-xenon headlight (option 522), the Adaptive Headlight (option 524) and automatic high-beams (option 5AC) are also offered as additional equipment.



F34/F30 xenon headlight

The headlight design of the F34 is slightly different from the headlight of the F30 as shown by the picture above.



Bayerische Motorenwerke Aktiengesellschaft
Qualifizierung und Training
Röntgenstraße 7
85716 Unterschleißheim, Germany